

Criblage DFT (WB97X-D4/aug-cc-pVTZ) des allotropes polyazotés N_x : rapport d'étape

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Abstract

Ce document rend compte de la campagne DFT en cours au niveau WB97X-D4/aug-cc-pVTZ sur les **193 candidats** présélectionnés par criblage GFN2-xTB (N_4 à N_{16} , familles neutre/cation/anion). État au moment de la rédaction : **120 structures optimisées et vérifiées par calcul de fréquences** (62 %), 0 erreur ORCA, campagne active sur cluster SLURM. Un contrôle systématique de connectivité (§2) a par ailleurs révélé que 13 structures terminées sont en réalité **fragmentées** (deux espèces moléculaires distinctes plutôt qu'un cluster lié, ex. $N_2 + N_3^-$) – exclues des comparaisons de stabilité ci-dessous. Une confrontation aux bases de données et à la littérature déjà compilées dans polyN-pipeline montre que, parmi les 7 groupes (formule, charge) où une comparaison est possible, le criblage local a déjà trouvé un isomère non-fragmenté plus stable que tout ce qui est catalogué dans 3 d'entre eux (N_9^- , N_{10} , N_{12}), tandis que 165 des 193 candidats (86 %) ne correspondent à aucune topologie référencée ailleurs. Les deux ancres de référence pour les ΔH_f ioniques (N_5^- cyclique, N_5^+ coudé) ont été retrouvées dans `biblio_polyN/final_report_isodesmic.csv` et permettent désormais de calculer un ΔH_f absolu (niveau GFN2-xTB) pour les 193 candidats – tables 2–4.

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1. Méthode

1.1 Origine des candidats

Les 193 structures de départ proviennent du criblage GFN2-xTB de polyN-pipeline (génération RDKit ETKDGv3 + énumération de graphes *geng/nauty* + substitution isolobale C→N + échantillonnage aléatoire, selon la formule), sélection par stoechiométrie et vérification de fréquences à ce niveau. Aucune contribution du second dépôt compagnon polyN_adapt (génération adaptative/surrogate) n'a été identifiée dans ce jeu – voir §2, à confirmer. Les formules à N impair (N_5 , N_7 , N_9 , N_{11} , N_{13} , N_{15}) ne sont représentées qu'en cation/anion (contrainte de parité électronique), les formules paires principalement en neutre.

1.2 Niveau de calcul

Optimisation de géométrie puis fréquences, WB97X-D4/aug-cc-pVTZ, TightSCF, DEFGRID3 (ORCA 6.1.0). Deux protocoles selon la taille :

- **N_4 – N_{11}** (« petits ») : un job unique Opt+Freq, sans contrainte de symétrie – évite tout risque de figer un candidat Jahn-Teller dans un point-selle de symétrie artificiellement élevée.
- **N_{12} – N_{16}** (« gros ») : chaîne en trois étapes Opt (sans symétrie) → ré-optimisation rapide avec UseSym (nettoie la géométrie vers le groupe ponctuel exact) → Freq avec UseSym (diagonalisation du Hessien exploitant la symétrie complète, pas seulement les sous-groupes de D_{2h}). Surcoût mesuré d'un facteur ~ 2 en temps de calcul sur un cas test (N_5^- , 7:41 → 15:35), jugé acceptable uniquement pour les grosses structures.

Distribution sur cluster SLURM via un ordonnanceur maison (*production_dispatch.py*) : surveillance en continu des cœurs libres sur les nœuds alloués, soumission automatique du prochain candidat dès qu'un créneau se libère, 8 cœurs par job (optimum mesuré : efficacité CPU 75–96 %, gain marginal au-delà ; 24–32 cœurs pour les gros tant que la file correspondante n'était pas épuisée, efficacité 90–96 % mesurée jusqu'à 32 cœurs sur N_{12}).

Catégorie	n	Wall moy. (min)	Wall min (min)	Wall max (min)	CPU-h moy.
Petits (8 cœurs, N_4 – N_{11})	83	72.4	5.1	262.9	8.88
Gros (24/32 cœurs, N_{12} – N_{16})	23	116.9	23.3	310.4	46.52
Total	106	82.1	5.1	310.4	17.05

Table 1: Temps de calcul (écoulé = wall-clock, CPU = temps utilisateur+système cumulé sur tous les cœurs) sur les 106 jobs terminés au moment de la rédaction.

2. Provenance et statut de chaque candidat

C'est la table centrale de ce rapport : pour chacun des 193 candidats, son origine, son statut DFT actuel, son ΔH_f GFN2-xTB (ancré sur les valeurs littérature CCSD(T) retrouvées dans *biblio_polyN/final_report_isodesmic.csv* – $-106,4$ kcal/mol pour le pentagone N_5^- (D_{5h}), $195,9$ kcal/mol pour le N_5^+ coudé/Vchain (C_{2v} , valeur expérimentale par diffraction X concordante à 195,8), $\Delta H_f(N_2) = 0$ pour les neutres par construction) et, quand une comparaison est possible, la topologie cataloguée à laquelle il correspond.

Les deux ancres ont été recalculées indépendamment ici (GFN2-xTB, xtb 6.7.1) et reproduisent les valeurs internes de `polyN_pipeline.py` à 5–6 décimales : $E(\text{N}_2) = -5,763935$ Ha, $E(\text{N}_5^-, \text{D}_{5h}, = \text{N5_anion_anion_001}) = -14,760899$ Ha, $E(\text{N}_5^+, \text{C}_{2v}, = \text{N5_cation_cation_001}) = -13,985267$ Ha. Point notable : la référence interne du pipeline pour la famille cation n'est pas ce dernier mais `N5\_cation\_cation\_004` (isomère C_1 différent, $E = -13,816935$ Ha, 105,6 kcal/mol moins stable) – les ΔH ci-dessous sont recalés sur le C_{2v} qui correspond à la structure littérature.

Trois tableaux distincts (neutres, cations, anions), triés par nombre d'atomes croissant puis par stabilité au sein de chaque formule ($\Delta H = 0$ pour l'isomère le plus stable, DFT quand disponible sinon GFN2-xTB). Chaque ligne étant un de nos 193 candidats, *our work* s'applique toujours ; quand la topologie correspond en plus à un article publié, son numéro est cité à côté (“[k], *our work*” – liste des articles en Annexe B). Sans correspondance article, seul *our work* apparaît – ce qui inclut, par choix explicite, les correspondances aux bases externes `N_csp` et `polyN_study` (candidats, pas des articles publiés) aussi bien que les topologies réellement inédites.

Attention sur la portée de cette bibliographie : les 22 références numérotées ici ([GS], [B], [A1]–[A20]) sont celles pour lesquelles `polyN-pipeline` a déjà extrait une structure et recalculé son ΔH_f isodesmique (110 structures nommées dans `table_isodesmic_{neutre,anion,cation}.tex`). Le dépôt contient séparément une recherche de citations bien plus large (135 articles en génération 1, `table_citation_search.tex + _gen2.tex`) qui n'est **pas encore** réduite en structures nommées avec référence – ces papiers supplémentaires ne peuvent donc pas être cités par candidat tant que ce travail d'extraction n'est pas fait. Une première passe de correction ici (priorité à une correspondance `biblio_article` existante plutôt qu'à un doublon de base externe de même topologie mais moindre énergie) a fait passer les correspondances de 1 à 8 candidats.

Table 2: Composés **neutres**, classés par nombre d'atomes croissant puis par stabilité (isomère le plus stable = $\Delta H = 0$). ΔH au niveau DFT quand disponible, sinon GFN2-xTB (indiqué en colonne). **Frag.** : la structure se sépare en plusieurs espèces moléculaires distinctes (distance N–N > 1,7 Å entre elles, ex. deux ions ou une paire N_2 +reste) plutôt que de former un cluster lié unique – exclue du calcul de $\Delta H = 0$ du groupe. Référence(s) : numéro(s) d'article (Annexe Références) ou *our work* si absente de la biblio actuelle.

Nom	N	PG	ΔH (kcal/mol)	niveau	Frag.	Réf.
N4_neutral_neutral_001	4	Td	0.00	DFT		our work
N4_neutral_neutral_004	4	D2h	3.38	DFT		our work
N4_neutral_neutral_002	4	C2v	16.85	DFT		our work
N4_neutral_neutral_003	4	C2v	33.07	DFT		our work
N6_neutral_neutral_004	6	C1	-172.76	DFT	2+2+2	our work

(suite)					
N6_neutral_neutral_001	6	C2	0.00	DFT	our work
N6_neutral_neutral_002	6	D2	23.07	DFT	[1], our work
N6_neutral_neutral_003	6	C1	28.07	DFT	our work
N6_neutral_neutral_005	6	C1	52.69	DFT	our work
N6_neutral_neutral_007	6	C2	53.36	DFT	our work
N6_neutral_neutral_009	6	C2	58.25	DFT	our work
N6_neutral_neutral_008	6	C2v	63.90	DFT	our work
N6_neutral_neutral_006	6	C1	85.49	DFT	our work
N6_neutral_neutral_011	6	—	119.57	xtb	our work
N6_neutral_neutral_012	6	D3h	132.46	DFT	our work
N6_neutral_neutral_010	6	C2v	149.79	DFT	our work
N8_neutral_neutral_002	8	C1	0.00	DFT	[1],[2], our work
N8_neutral_neutral_001	8	D2h	11.23	DFT	[1],[2], our work
N8_neutral_neutral_005	8	C2	18.41	DFT	[1], our work
N8_neutral_neutral_003	8	C1	18.53	DFT	[1], our work
N8_neutral_neutral_004	8	—	28.09	xtb	[1], our work
N8_neutral_neutral_006	8	—	56.53	xtb	our work
N8_neutral_neutral_007	8	—	66.38	xtb	our work
N8_neutral_neutral_008	8	—	67.22	xtb	our work
N8_neutral_neutral_009	8	D3h	68.56	DFT	our work
N8_neutral_neutral_010	8	Cs	76.55	DFT	our work
N8_neutral_neutral_011	8	—	84.76	xtb	our work
N8_neutral_neutral_012	8	—	93.97	xtb	our work
N8_neutral_neutral_013	8	—	102.90	xtb	our work

(suite)						
N8_neutral_neutral_014	8	–	113.23	xtb		our work
N8_neutral_neutral_019	8	D2d	114.22	DFT		our work
N8_neutral_neutral_016	8	Cs	122.30	DFT		our work
N8_neutral_neutral_015	8	–	126.26	xtb		our work
N8_neutral_neutral_017	8	–	139.04	xtb		our work
N8_neutral_neutral_018	8	–	140.21	xtb		our work
N8_neutral_neutral_020	8	–	152.93	xtb		our work
N8_neutral_neutral_021	8	–	155.66	xtb		our work
N8_neutral_neutral_022	8	–	161.85	xtb		our work
N8_neutral_neutral_023	8	–	164.68	xtb		our work
N8_neutral_neutral_024	8	D2h	178.42	DFT		our work
N8_neutral_neutral_025	8	–	205.70	xtb		our work
N10_neutral_neutral_001	10	C2	0.00	DFT		our work
N10_neutral_neutral_002	10	C1	30.61	DFT		our work
N10_neutral_neutral_003	10	C1	50.60	DFT		our work
N10_neutral_neutral_008	10	–	59.29	xtb	5+5	[1], our work
N10_neutral_neutral_009	10	–	63.47	xtb	3+7	our work
N10_neutral_neutral_006	10	Cs	69.56	DFT		our work
N10_neutral_neutral_004	10	C1	71.13	DFT		our work
N10_neutral_neutral_007	10	Cs	87.34	DFT		[1], our work
N10_neutral_neutral_012	10	–	98.78	xtb		our work
N10_neutral_neutral_005	10	C1	102.77	DFT		our work
N10_neutral_neutral_011	10	C1	107.23	DFT		our work
N10_neutral_neutral_010	10	C1	118.99	DFT		our work
N10_neutral_neutral_015	10	C2	123.40	DFT		our work

(suite)						
N10_neutral_neutral_016	10	C1	126.63	DFT		our work
N10_neutral_neutral_014	10	C1	139.29	DFT		our work
N10_neutral_neutral_017	10	C2v	151.14	DFT		our work
N10_neutral_neutral_021	10	C2v	173.84	DFT		our work
N10_neutral_neutral_013	10	C2v	193.75	DFT		our work
N10_neutral_neutral_018	10	C2	199.07	DFT		our work
N10_neutral_neutral_019	10	C3v	203.80	DFT		our work
N10_neutral_neutral_020	10	C2	214.21	DFT		our work
N10_neutral_neutral_022	10	D5h	241.14	DFT		our work
N12_neutral_neutral_001	12	C1	0.00	DFT		our work
N12_neutral_neutral_003	12	C1	36.41	DFT		our work
N12_neutral_neutral_002	12	C1	56.99	DFT		our work
N12_neutral_neutral_004	12	Cs	116.12	DFT		our work
N12_neutral_neutral_006	12	C1	127.50	DFT		our work
N12_neutral_neutral_005	12	C1	138.12	DFT		our work
N12_neutral_neutral_007	12	D3d	188.33	DFT		our work
N12_neutral_neutral_008	12	C1	219.30	DFT		our work
N12_neutral_neutral_009	12	D6h	334.92	DFT		our work
N14_neutral_neutral_004	14	C1	-250.59	DFT	2+2+2+3+5	our work
N14_neutral_neutral_001	14	C1	0.00	DFT		our work
N14_neutral_neutral_003	14	C2v	17.64	DFT		our work
N14_neutral_neutral_002	14	–	40.76	xtb		our work
N14_neutral_neutral_005	14	C2	90.77	DFT		our work
N16_neutral_neutral_001	16	–	0.00	xtb		our work

(suite)						
N16_neutral_neutral_002	16	–	36.70	xtb		our work

Table 3: Composés **cationiques**, classés par nombre d'atomes croissant puis par stabilité (isomère le plus stable = $\Delta H = 0$). ΔH au niveau DFT quand disponible, sinon GFN2-xTB (indiqué en colonne). **Frag.** : la structure se sépare en plusieurs espèces moléculaires distinctes (distance N–N > 1,7 Å entre elles, ex. deux ions ou une paire N₂+reste) plutôt que de former un cluster lié unique – exclue du calcul de $\Delta H = 0$ du groupe. Référence(s) : numéro(s) d'article (Annexe Références) ou *our work* si absente de la biblio actuelle.

Nom	N	PG	ΔH (kcal/mol)	niveau	Frag.	Réf.
N5_cation_cation_003	5	C1	0.00	DFT		our work
N5_cation_cation_001	5	C2v	0.00	DFT		our work
N5_cation_cation_004	5	C1	0.01	DFT		our work
N5_cation_cation_007	5	C1	34.45	DFT	2+3	our work
N5_cation_cation_002	5	Cs	41.63	DFT		our work
N5_cation_cation_005	5	D2d	109.53	DFT		our work
N5_cation_cation_006	5	C4v	155.74	DFT		our work
N7_cation_cation_001	7	Cs	0.00	DFT		our work
N7_cation_cation_002	7	–	6.20	xtb		our work
N7_cation_cation_003	7	Cs	25.42	DFT		our work
N7_cation_cation_004	7	–	91.26	xtb		our work
N7_cation_cation_005	7	Cs	189.69	DFT		our work
N7_cation_cation_006	7	C1	200.56	DFT		our work
N9_cation_cation_001	9	C1	-29.44	xtb	2+2+5	our work
N9_cation_cation_002	9	–	0.00	xtb		our work
N9_cation_cation_003	9	–	2.53	xtb		our work

(suite)						
N9_cation_cation_004	9	–	4.76	xtb		our
N9_cation_cation_005	9	–	71.16	xtb		work
N9_cation_cation_006	9	–	75.10	xtb		our
N9_cation_cation_007	9	–	84.43	xtb		work
N9_cation_cation_008	9	–	85.45	xtb		our
N9_cation_cation_009	9	–	87.91	xtb		work
N9_cation_cation_010	9	–	102.87	xtb		our
N9_cation_cation_011	9	–	144.72	xtb		work
N9_cation_cation_012	9	–	147.41	xtb		our
						work
N11_cation_cation_002	11	C1	0.00	DFT		our
N11_cation_cation_001	11	C2v	23.30	DFT		work
N11_cation_cation_005	11	C2	45.85	DFT		our
N11_cation_cation_003	11	–	46.08	xtb		work
N11_cation_cation_004	11	C1	56.05	DFT		our
N11_cation_cation_006	11	C1	67.68	DFT		work
N11_cation_cation_007	11	C1	118.69	DFT		our
N11_cation_cation_008	11	–	195.40	xtb		work
N11_cation_cation_009	11	–	239.73	xtb		our
N11_cation_cation_010	11	–	239.73	xtb		work
						our
N12_cation_cation_001	12	C1	0.00	DFT		work
N13_cation_cation_001	13	C1	0.00	DFT		our
N13_cation_cation_002	13	–	40.46	xtb		work
N13_cation_cation_003	13	–	56.53	xtb		our
N13_cation_cation_004	13	C1	151.22	DFT		work
						our
						work

(suite)						
N15_cation_cation_001	15	C1	0.00	DFT		our work
N15_cation_cation_002	15	C1	327.87	DFT		our work

Table 4: Composés **anioniques**, classés par nombre d'atomes croissant puis par stabilité (isomère le plus stable = $\Delta H = 0$). ΔH au niveau DFT quand disponible, sinon GFN2-xTB (indiqué en colonne). **Frag.** : la structure se sépare en plusieurs espèces moléculaires distinctes (distance N–N > 1,7 Å entre elles, ex. deux ions ou une paire N₂+reste) plutôt que de former un cluster lié unique – exclue du calcul de $\Delta H = 0$ du groupe. Référence(s) : numéro(s) d'article (Annexe Références) ou *our work* si absente de la biblio actuelle.

Nom	N	PG	ΔH (kcal/mol)	niveau	Frag.	Réf.
N5_anion_anion_002	5	C1	-6.58	DFT	2+3	our work
N5_anion_anion_003	5	C1	-6.58	DFT	2+3	our work
N5_anion_anion_004	5	C1	-6.58	DFT	2+3	our work
N5_anion_anion_001	5	D5h	0.00	DFT		our work
N5_anion_anion_007	5	C1	63.83	DFT	2+3	our work
N5_anion_anion_005	5	C1	63.83	DFT	2+3	our work
N5_anion_anion_008	5	C1	65.57	DFT	2+3	our work
N5_anion_anion_009	5	C1	66.07	DFT	2+3	our work
N5_anion_anion_006	5	Cs	104.46	DFT		our work
N5_anion_anion_010	5	C2v	170.76	DFT		our work
N7_anion_anion_001	7	C1	0.00	DFT		our work
N7_anion_anion_002	7	–	2.77	xtb		our work
N7_anion_anion_004	7	–	9.12	xtb		our work
N7_anion_anion_013	7	C1	9.42	DFT		our work
N7_anion_anion_003	7	C2v	20.91	DFT		our work

(suite)						
N7_anion_anion_005	7	–	22.23	xtb		our work
N7_anion_anion_011	7	C1	26.27	DFT		our work
N7_anion_anion_012	7	C1	29.35	DFT		our work
N7_anion_anion_006	7	–	30.87	xtb	2+5	our work
N7_anion_anion_007	7	Cs	35.75	DFT		our work
N7_anion_anion_016	7	C1	40.69	DFT		our work
N7_anion_anion_010	7	C2v	46.25	DFT		our work
N7_anion_anion_008	7	C1	49.34	DFT		our work
N7_anion_anion_009	7	C1	54.95	DFT		our work
N7_anion_anion_014	7	Cs	55.87	DFT		our work
N7_anion_anion_015	7	–	71.61	xtb		our work
N7_anion_anion_018	7	Cs	79.33	DFT		our work
N7_anion_anion_017	7	Cs	80.80	DFT		our work
N7_anion_anion_019	7	C1	89.71	DFT		our work
N7_anion_anion_020	7	C1	179.57	DFT		our work
N9_anion_anion_001	9	C1	0.00	DFT		our work
N9_anion_anion_002	9	–	19.37	xtb		our work
N9_anion_anion_003	9	–	23.62	xtb		our work
N9_anion_anion_006	9	Cs	27.56	DFT		our work
N9_anion_anion_004	9	–	29.47	xtb		our work
N9_anion_anion_005	9	–	33.08	xtb		our work
N9_anion_anion_007	9	–	46.38	xtb	3+6	our work
N9_anion_anion_008	9	–	53.91	xtb		our work
N9_anion_anion_009	9	–	55.92	xtb	2+7	our work
N9_anion_anion_010	9	–	57.07	xtb		our work

(suite)					
N9_anion_anion_011	9	–	60.00	xtb	our work
N9_anion_anion_012	9	–	86.79	xtb	our work
N9_anion_anion_013	9	–	87.40	xtb	our work
N9_anion_anion_014	9	–	88.41	xtb	our work
N9_anion_anion_015	9	–	104.99	xtb	our work
N9_anion_anion_016	9	–	126.07	xtb	our work
N9_anion_anion_017	9	–	128.05	xtb	our work
N9_anion_anion_018	9	–	141.31	xtb	our work
N9_anion_anion_019	9	–	153.96	xtb	our work
N9_anion_anion_020	9	–	205.40	xtb	2+7 our work
N11_anion_anion_001	11	C1	0.00	DFT	our work
N11_anion_anion_002	11	C1	38.48	DFT	our work
N11_anion_anion_006	11	C1	60.32	DFT	our work
N11_anion_anion_011	11	C1	74.02	DFT	3+3+5 our work
N11_anion_anion_005	11	C1	74.39	DFT	our work
N11_anion_anion_009	11	–	79.30	xtb	our work
N11_anion_anion_004	11	C1	92.16	DFT	our work
N11_anion_anion_007	11	Cs	101.73	DFT	our work
N11_anion_anion_010	11	Cs	102.27	DFT	our work
N11_anion_anion_008	11	C1	108.61	DFT	our work
N11_anion_anion_012	11	–	108.72	xtb	our work
N11_anion_anion_003	11	C2	109.19	DFT	our work
N11_anion_anion_014	11	–	119.70	xtb	our work
N11_anion_anion_015	11	–	146.04	xtb	our work
N11_anion_anion_016	11	–	164.02	xtb	our work

(suite)						
N11_anion_anion_013	11	C1	177.01	DFT		our work
N11_anion_anion_017	11	C1	185.47	DFT		our work
N11_anion_anion_018	11	C1	199.63	DFT	3+8	our work
N13_anion_anion_001	13	C1	0.00	DFT		our work
N13_anion_anion_002	13	–	115.00	xtb		our work
N13_anion_anion_003	13	–	116.61	xtb		our work
N13_anion_anion_004	13	–	123.73	xtb		our work

3. Classement de stabilité : DFT contre xTB

Les tableaux suivants classent chaque isomère par rang DFT croissant (rang 1 = ground state DFT) au sein de sa formule, avec le rang xTB d'origine rappelé en regard – **la question posée est de savoir si le criblage xTB, nettement moins cher, prédit le même classement de stabilité que le DFT**, ou si des réordonnements significatifs apparaissent.

Table 5: Classement DFT vs xTB, composés **neutres**. Trié par rang DFT (ground state DFT = rang 1) au sein de chaque formule ; le rang xTB tel que généré à l'origine est rappelé pour voir si les deux classements concordent.

Nom	N	rang xtb	rang DFT	E_{rel} xtb (kcal/mol)	E_{rel} DFT (kcal/mol)	accord GS
N4_neutral_neutral_001	4	1	1	0.0	0.0	
N4_neutral_neutral_004	4	4	2	41.64	3.375	
N4_neutral_neutral_002	4	2	3	26.47	16.846	
N4_neutral_neutral_003	4	3	4	31.01	33.074	
N6_neutral_neutral_001	6	1	1	0.0	0.0	
N6_neutral_neutral_002	6	2	2	24.09	23.072	
N6_neutral_neutral_003	6	3	3	55.0	28.065	
N6_neutral_neutral_005	6	5	4	64.62	52.687	
N6_neutral_neutral_007	6	7	5	83.63	53.362	
N6_neutral_neutral_009	6	9	6	110.8	58.246	
N6_neutral_neutral_008	6	8	7	84.52	63.9	
N6_neutral_neutral_006	6	6	8	78.8	85.49	
N6_neutral_neutral_012	6	12	9	136.66	132.463	
N6_neutral_neutral_010	6	10	10	114.33	149.792	
N8_neutral_neutral_002	8	2	1	9.91	0.0	
N8_neutral_neutral_001	8	1	2	0.0	11.226	
N8_neutral_neutral_005	8	5	3	29.41	18.413	
N8_neutral_neutral_003	8	3	4	20.26	18.532	

(suite)

Nom	N	rang xtb	rang DFT	E_{rel} xtb	E_{rel} DFT	accord GS
N8_neutral_neutral_009	8	9	5	73.7	68.563	
N8_neutral_neutral_010	8	10	6	84.06	76.549	
N8_neutral_neutral_019	8	19	7	150.3	114.216	
N8_neutral_neutral_016	8	16	8	132.78	122.296	
N8_neutral_neutral_024	8	24	9	183.05	178.416	
N10_neutral_neutral_001	10	1	1	0.0	0.0	
N10_neutral_neutral_002	10	2	2	10.5	30.609	
N10_neutral_neutral_003	10	3	3	23.5	50.598	
N10_neutral_neutral_006	10	6	4	38.56	69.561	
N10_neutral_neutral_004	10	4	5	33.36	71.134	
N10_neutral_neutral_007	10	7	6	53.65	87.344	
N10_neutral_neutral_005	10	5	7	36.56	102.766	
N10_neutral_neutral_011	10	11	8	92.95	107.23	
N10_neutral_neutral_010	10	10	9	88.62	118.993	
N10_neutral_neutral_015	10	15	10	111.24	123.395	
N10_neutral_neutral_016	10	16	11	113.83	126.633	
N10_neutral_neutral_014	10	14	12	100.5	139.293	
N10_neutral_neutral_017	10	17	13	130.46	151.139	
N10_neutral_neutral_021	10	21	14	174.6	173.835	
N10_neutral_neutral_013	10	13	15	99.55	193.748	
N10_neutral_neutral_018	10	18	16	143.02	199.068	
N10_neutral_neutral_019	10	19	17	147.97	203.805	
N10_neutral_neutral_020	10	20	18	168.46	214.208	
N10_neutral_neutral_022	10	22	19	183.94	241.136	
N12_neutral_neutral_001	12	1	1	0.0	0.0	
N12_neutral_neutral_003	12	3	2	35.91	36.408	
N12_neutral_neutral_002	12	2	3	13.24	56.987	
N12_neutral_neutral_004	12	4	4	45.58	116.12	
N12_neutral_neutral_006	12	6	5	76.36	127.5	
N12_neutral_neutral_005	12	5	6	68.53	138.121	
N12_neutral_neutral_007	12	7	7	100.96	188.328	
N12_neutral_neutral_008	12	8	8	133.16	219.304	
N12_neutral_neutral_009	12	9	9	242.43	334.919	
N14_neutral_neutral_001	14	1	1	0.0	0.0	
N14_neutral_neutral_003	14	3	2	77.89	17.638	
N14_neutral_neutral_005	14	5	3	133.34	90.768	

Table 6: Classement DFT vs xTB, composés **cationiques**.

Trié par rang DFT (ground state DFT = rang 1) au sein de chaque formule ; le rang xTB tel que généré à l'origine est rappelé pour voir si les deux classements concordent.

Nom	N	rang xtb	rang DFT	E_{rel} xtb (kcal/mol)	E_{rel} DFT (kcal/mol)	accord GS
N5_cation_cation_003	5	3	1	104.85	0.0	

(suite)

Nom	N	rang xtb	rang DFT	E_{rel} xtb	E_{rel} DFT	accord GS
N5_cation_cation_001	5	1	2	0.0	0.001	
N5_cation_cation_004	5	4	3	105.63	0.007	
N5_cation_cation_002	5	2	4	74.16	41.634	
N5_cation_cation_005	5	5	5	134.0	109.535	
N5_cation_cation_006	5	6	6	184.33	155.738	
N7_cation_cation_001	7	1	1	0.0	0.0	
N7_cation_cation_003	7	3	2	27.01	25.419	
N7_cation_cation_005	7	5	3	124.94	189.693	
N7_cation_cation_006	7	6	4	178.83	200.561	
N11_cation_cation_002	11	2	1	2.67	0.0	
N11_cation_cation_001	11	1	2	0.0	23.301	
N11_cation_cation_005	11	5	3	73.62	45.845	
N11_cation_cation_004	11	4	4	61.37	56.05	
N11_cation_cation_006	11	6	5	100.27	67.685	
N11_cation_cation_007	11	7	6	128.07	118.687	
N12_cation_cation_001	12	1	1	0.0	0.0	
N13_cation_cation_001	13	1	1	0.0	0.0	
N13_cation_cation_004	13	4	2	194.1	151.221	
N15_cation_cation_001	15	1	1	0.0	0.0	
N15_cation_cation_002	15	2	2	259.27	327.866	

Table 7: Classement DFT vs xTB, composés **anioniques**.

Trié par rang DFT (ground state DFT = rang 1) au sein de chaque formule ; le rang xTB tel que généré à l'origine est rappelé pour voir si les deux classements concordent.

Nom	N	rang xtb	rang DFT	E_{rel} xtb (kcal/mol)	E_{rel} DFT (kcal/mol)	accord GS
N5_anion_anion_001	5	1	1	0.0	0.0	
N5_anion_anion_006	5	6	2	104.25	104.46	
N5_anion_anion_010	5	10	3	143.36	170.763	
N7_anion_anion_001	7	1	1	0.0	0.0	
N7_anion_anion_013	7	13	2	54.11	9.422	
N7_anion_anion_003	7	3	3	7.94	20.912	
N7_anion_anion_011	7	11	4	51.7	26.265	
N7_anion_anion_012	7	12	5	52.34	29.347	
N7_anion_anion_007	7	7	6	41.65	35.754	
N7_anion_anion_016	7	16	7	74.29	40.69	
N7_anion_anion_010	7	10	8	50.4	46.246	
N7_anion_anion_008	7	8	9	46.21	49.337	
N7_anion_anion_009	7	9	10	47.46	54.955	
N7_anion_anion_014	7	14	11	59.93	55.871	
N7_anion_anion_018	7	18	12	96.97	79.33	
N7_anion_anion_017	7	17	13	82.01	80.797	

(suite)

Nom	N	rang xtb	rang DFT	E_{rel} xtb	E_{rel} DFT	accord GS
N7_anion_anion_019	7	19	14	102.35	89.71	
N7_anion_anion_020	7	20	15	144.93	179.573	
N9_anion_anion_001	9	1	1	0.0	0.0	
N9_anion_anion_006	9	6	2	35.95	27.555	
N11_anion_anion_001	11	1	1	0.0	0.0	
N11_anion_anion_002	11	2	2	39.44	38.478	
N11_anion_anion_006	11	6	3	57.42	60.325	
N11_anion_anion_005	11	5	4	51.65	74.395	
N11_anion_anion_004	11	4	5	49.83	92.156	
N11_anion_anion_007	11	7	6	69.82	101.735	
N11_anion_anion_010	11	10	7	89.5	102.267	
N11_anion_anion_008	11	8	8	77.42	108.607	
N11_anion_anion_003	11	3	9	44.22	109.185	
N11_anion_anion_013	11	13	10	115.91	177.005	
N11_anion_anion_017	11	17	11	164.9	185.468	
N13_anion_anion_001	13	1	1	0.0	0.0	

4. Vérification des minima

Sur les 92 structures terminées avec un bloc de thermochimie complet, **19 présentent au moins une fréquence imaginaire** et ne sont donc pas, en l'état, de vrais minima locaux – concentrées de manière frappante sur N_{11}^- et N_{14} neutre. Contrairement au protocole gros-cluster, aucune redésymétrisation automatique n'est appliquée : chaque candidat flaggé devra être repris manuellement (distorsion le long du mode imaginaire puis réoptimisation) avant d'être retenu comme candidat final – exactement la logique de la colonne `is_true_minimum / n_freq_hops` déjà implémentée dans `polyN_pipeline.py` au niveau GFN2-xTB, appliquée ici au niveau DFT. Le détail (fréquence, intensité IR) est dans `results_imaginary_modes.csv`.

5. Orbitales frontières

Table 8: Orbitales frontières (WB97X-D4/aug-cc-pVTZ) des candidats terminés.

Nom	PG	HOMO (eV)	LUMO (eV)	Gap (eV)
N10_neutral_neutral_001	C2	-12.9198	-0.5398	12.3800
N10_neutral_neutral_002	C1	-11.7475	-1.3053	10.4422
N10_neutral_neutral_003	C1	-9.9826	-1.1691	8.8135
N10_neutral_neutral_004	C1	-9.7070	-1.4443	8.2627
N10_neutral_neutral_005	C1	-10.3189	-1.6633	8.6556
N10_neutral_neutral_006	Cs	-11.2573	-0.6247	10.6326
N10_neutral_neutral_007	Cs	-10.5916	-1.8512	8.7404
N10_neutral_neutral_010	C1	-11.1858	-0.6563	10.5295
N10_neutral_neutral_011	C1	-10.2546	-1.8095	8.4451
N10_neutral_neutral_013	C2v	-8.7777	-2.5200	6.2577
N10_neutral_neutral_014	C1	-10.1760	-3.1308	7.0452

(suite)				
Nom	PG	HOMO (eV)	LUMO (eV)	Gap (eV)
N10_neutral_neutral_015	C2	-10.7038	-1.6499	9.0539
N10_neutral_neutral_016	C1	-10.7642	-1.5672	9.1970
N10_neutral_neutral_017	C2v	-11.2487	-1.3285	9.9202
N10_neutral_neutral_018	C2	-11.1371	0.6545	11.7916
N10_neutral_neutral_019	C3v	-11.3917	0.3446	11.7363
N10_neutral_neutral_020	C2	-10.5955	-1.4418	9.1537
N10_neutral_neutral_021	C2v	-11.2774	-1.0998	10.1776
N10_neutral_neutral_022	D5h	-11.4974	-0.8667	10.6307
N11_anion_anion_001	C1	-4.6868	4.7575	9.4443
N11_anion_anion_002	C1	-4.6917	4.4152	9.1069
N11_anion_anion_003	C2	-4.8000	3.0674	7.8674
N11_anion_anion_004	C1	-4.1785	4.4585	8.6370
N11_anion_anion_005	C1	-4.0097	4.1648	8.1745
N11_anion_anion_006	C1	-4.2089	4.0149	8.2238
N11_anion_anion_007	Cs	-4.6404	4.6470	9.2874
N11_anion_anion_008	C1	-3.9921	4.4067	8.3988
N11_anion_anion_010	Cs	-3.3157	4.1356	7.4513
N11_anion_anion_011	C1	-4.8669	2.8826	7.7495
N11_anion_anion_013	C1	-2.7734	2.6705	5.4439
N11_anion_anion_017	C1	-4.0058	4.4202	8.4260
N11_anion_anion_018	C1	-2.4542	3.0647	5.5189
N11_cation_cation_001	C2v	-16.4400	-6.4750	9.9650
N11_cation_cation_002	C1	-15.7017	-6.1480	9.5537
N11_cation_cation_004	C1	-16.4230	-7.7299	8.6931
N11_cation_cation_005	C2	-17.0074	-7.0585	9.9489
N11_cation_cation_006	C1	-15.6724	-7.1994	8.4730
N11_cation_cation_007	C1	-16.4756	-7.3306	9.1450
N12_neutral_neutral_001	C1	-12.3196	-2.4745	9.8451
N12_neutral_neutral_002	C1	-11.3338	-2.5898	8.7440
N12_neutral_neutral_003	C1	-11.9678	-1.5248	10.4430
N12_neutral_neutral_004	Cs	-9.8374	-2.7778	7.0596
N12_neutral_neutral_005	C1	-10.6336	-1.7561	8.8775
N12_neutral_neutral_006	C1	-10.3193	-1.8368	8.4825
N12_neutral_neutral_007	D3d	-11.6168	-0.4953	11.1215
N12_neutral_neutral_008	C1	-10.9868	-0.5129	10.4739
N12_neutral_neutral_009	D6h	-10.5865	-0.9025	9.6840
N13_anion_anion_001	C1	-4.2327	3.0401	7.2728
N13_cation_cation_001	C1	-15.9024	-7.4824	8.4200
N13_cation_cation_004	C1	-17.1447	-7.0618	10.0829
N14_neutral_neutral_001	C1	-9.4477	-2.1106	7.3371
N14_neutral_neutral_003	C2v	-10.3293	-2.3248	8.0045
N14_neutral_neutral_004	C1	-10.0443	-1.2482	8.7961
N14_neutral_neutral_005	C2	-12.0686	0.6111	12.6797
N15_cation_cation_001	C1	-16.9221	-7.4624	9.4597
N15_cation_cation_002	C1	-17.0792	-5.2422	11.8370
N4_neutral_neutral_001	Td	-13.9405	1.7935	15.7340

(suite)

Nom	PG	HOMO (eV)	LUMO (eV)	Gap (eV)
N4_neutral_neutral_002	C2v	-9.2289	-1.2933	7.9356
N4_neutral_neutral_003	C2v	-10.5771	-1.9734	8.6037
N4_neutral_neutral_004	D2h	-10.6441	-2.3507	8.2934
N5_anion_anion_001	D5h	-5.8838	5.8881	11.7719
N5_anion_anion_002	C1	-2.7204	5.1388	7.8592
N5_anion_anion_003	C1	-2.7207	5.1392	7.8599
N5_anion_anion_004	C1	-2.7202	5.1384	7.8586
N5_anion_anion_005	C1	-1.6987	5.2393	6.9380
N5_anion_anion_006	Cs	-1.8642	3.7995	5.6637
N5_anion_anion_007	C1	-1.6984	5.2388	6.9372
N5_anion_anion_008	C1	-1.5258	4.9909	6.5167
N5_anion_anion_009	C1	-1.5196	4.9170	6.4366
N5_anion_anion_010	C2v	-1.9078	5.4320	7.3398
N5_cation_cation_001	C2v	-18.4486	-6.5903	11.8583
N5_cation_cation_002	Cs	-19.4731	-7.2442	12.2289
N5_cation_cation_003	C1	-18.4487	-6.5897	11.8590
N5_cation_cation_004	C1	-18.4488	-6.5881	11.8607
N5_cation_cation_005	D2d	-19.7750	-8.2030	11.5720
N5_cation_cation_006	C4v	-20.6577	-7.9402	12.7175
N5_cation_cation_007	C1	-18.8382	-10.4353	8.4029
N6_neutral_neutral_001	C2	-9.2804	-0.0404	9.2400
N6_neutral_neutral_002	D2	-11.4121	-2.0522	9.3599
N6_neutral_neutral_003	C1	-10.1854	-0.1470	10.0384
N6_neutral_neutral_004	C1	-14.8605	1.5862	16.4467
N6_neutral_neutral_005	C1	-10.0047	-1.3166	8.6881
N6_neutral_neutral_006	C1	-9.3271	-0.6904	8.6367
N6_neutral_neutral_007	C2	-12.1087	-0.8535	11.2552
N6_neutral_neutral_008	C2v	-12.0044	-0.7828	11.2216
N6_neutral_neutral_009	C2	-11.2453	-0.2179	11.0274
N6_neutral_neutral_010	C2v	-9.7889	-3.6564	6.1325
N6_neutral_neutral_012	D3h	-11.9779	0.1531	12.1310
N7_anion_anion_001	C1	-1.9595	4.7108	6.6703
N7_anion_anion_003	C2v	-3.1498	5.2806	8.4304
N7_anion_anion_007	Cs	-2.2941	4.1099	6.4040
N7_anion_anion_008	C1	-2.9901	4.4265	7.4166
N7_anion_anion_009	C1	-3.0117	4.5488	7.5605
N7_anion_anion_010	C2v	-4.2100	5.1843	9.3943
N7_anion_anion_011	C1	-2.2849	4.8177	7.1026
N7_anion_anion_012	C1	-2.2126	4.8717	7.0843
N7_anion_anion_013	C1	-3.7378	4.5618	8.2996
N7_anion_anion_014	Cs	-3.7877	5.0235	8.8112
N7_anion_anion_016	C1	-3.5935	4.2032	7.7967
N7_anion_anion_017	Cs	-4.3231	4.7093	9.0324
N7_anion_anion_018	Cs	-2.9551	4.2456	7.2007
N7_anion_anion_019	C1	-2.6908	5.1080	7.7988
N7_anion_anion_020	C1	-3.1082	2.6928	5.8010
N7_cation_cation_001	Cs	-16.8094	-7.0687	9.7407

(suite)

Nom	PG	HOMO (eV)	LUMO (eV)	Gap (eV)
N7_cation_cation_003	Cs	-18.3210	-7.6492	10.6718
N7_cation_cation_005	Cs	-17.9319	-10.5556	7.3763
N7_cation_cation_006	C1	-19.0163	-9.2300	9.7863
N8_neutral_neutral_001	D2h	-11.2755	-0.0572	11.2183
N8_neutral_neutral_002	C1	-11.0090	-0.5306	10.4784
N8_neutral_neutral_003	C1	-9.8865	0.0206	9.9071
N8_neutral_neutral_005	C2	-9.6080	-0.0360	9.5720
N8_neutral_neutral_009	D3h	-11.7827	-1.2276	10.5551
N8_neutral_neutral_010	Cs	-9.8734	-1.8161	8.0573
N8_neutral_neutral_016	Cs	-11.5952	-0.6090	10.9862
N8_neutral_neutral_019	D2d	-12.0840	-1.8861	10.1979
N8_neutral_neutral_024	D2h	-9.2373	-1.2856	7.9517
N9_anion_anion_001	C1	-3.3287	4.4403	7.7690
N9_anion_anion_006	Cs	-4.5780	4.6285	9.2065
N9_cation_cation_001	C1	-18.1169	-6.2607	11.8562

6. Charges atomiques

Charges de Mulliken et Löwdin, natives à ORCA. Les charges de Bader (QTAIM) et de Manz (DDEC6) restent reportées (§9) – ni l'une ni l'autre n'est native à ORCA.

Table 9: Charges atomiques (Mulliken et Löwdin) du meilleur minimum confirmé de chaque groupe (formule, charge) – min/max sur les atomes N. Charges de Bader/QTAIM non natives à ORCA, reportées (§9).

Nom	Mulliken min	Mulliken max	Löwdin min/max
N4_neutral_neutral_001	-0.000	0.000	-0.000 / 0.000
N5_anion_anion_002	-0.879	0.791	-0.792 / -0.019
N5_cation_cation_003	-0.409	0.929	-0.069 / 0.465
N6_neutral_neutral_001	-0.552	1.088	-0.280 / 0.279
N7_anion_anion_001	-0.853	0.991	-0.388 / 0.103
N7_cation_cation_001	-0.296	1.030	-0.060 / 0.447
N8_neutral_neutral_002	-0.640	1.174	-0.450 / 0.341
N9_anion_anion_001	-0.818	1.443	-0.568 / 0.159
N9_cation_cation_001	-0.501	1.052	-0.105 / 0.448
N10_neutral_neutral_001	-0.282	0.818	-0.303 / 0.133
N11_anion_anion_006	-0.756	1.131	-0.375 / 0.197
N11_cation_cation_001	-0.656	1.068	-0.235 / 0.410
N12_cation_cation_001	-0.169	1.074	-0.269 / 0.207
N12_neutral_neutral_001	-0.319	1.408	-0.380 / 0.154
N13_anion_anion_001	-0.979	1.392	-0.394 / 0.224
N13_cation_cation_001	-0.293	0.835	-0.181 / 0.218
N14_neutral_neutral_003	-0.072	0.193	-0.130 / 0.094
N15_cation_cation_001	-0.422	0.873	-0.258 / 0.419

7. Distances et ordres de liaison N–N : corrélation avec ΔH_f

Pour les composés neutres, proportion de liaisons courtes (triple, double, ou délocalisées $\sim 1,5$) parmi les liaisons réelles, et ordre de Mayer moyen, en regard du ΔH_f GFN2-xTB. Tendance attendue : davantage de caractère multiple devrait corrélérer avec un ΔH_f plus élevé (structures plus contraintes, moins proches de 2 N₂) – à confirmer statistiquement une fois la campagne DFT complète (l'échantillon actuel, 45 structures neutres, montre la tendance qualitativement mais reste bruité).

Table 10: Composés **neutres** terminés : proportion de liaisons N–N courtes (triple/double/délocalisées) parmi les liaisons réelles (hors contacts non-liants), et ΔH_f GFN2-xTB. Seuils de classification en §9 de `harvest_results.py`.

Nom	liaisons	BO Mayer moy.	% courtes ¹	classes présentes	ΔH_f^{xtb}
N10_neutral_neutral_001	11	0.806	36%	delocalized 1.5, single	-134.2
N10_neutral_neutral_002	10	0.917	70%	delocalized 1.5, single, triple	-123.7
N10_neutral_neutral_003	9	1.071	67%	delocalized 1.5, single, triple	-110.7
N10_neutral_neutral_004	10	0.829	60%	delocalized 1.5, single, triple	-100.84
N10_neutral_neutral_005	10	0.937	50%	delocalized 1.5, double, single	-97.64
N10_neutral_neutral_006	12	0.792	25%	double, single	-95.64
N10_neutral_neutral_007	11	0.790	36%	delocalized 1.5, single	-80.55
N10_neutral_neutral_010	12	0.772	25%	delocalized 1.5, single	-45.58
N10_neutral_neutral_011	11	0.787	45%	delocalized 1.5, double, single	-41.25
N10_neutral_neutral_013	12	0.643	33%	delocalized 1.5, single	-34.65
N10_neutral_neutral_014	12	0.713	42%	delocalized 1.5, long/weak, single	-33.7
N10_neutral_neutral_015	12	0.768	25%	delocalized 1.5, single	-22.96
N10_neutral_neutral_016	12	0.760	25%	delocalized 1.5, double, single	-20.37
N10_neutral_neutral_017	13	0.733	15%	double, single	-3.74
N10_neutral_neutral_018	15	0.658	0%	single	8.82

(suite)

Nom	liaisons	BO Mayer moy.	% courtes	classes présentes	ΔH_f^{xtb}
N10_neutral_neutral_019	15	0.666	0%	single	13.77
N10_neutral_neutral_020	14	0.671	7%	delocalized	34.26
N10_neutral_neutral_021	13	0.781	15%	1.5, single delocalized	40.4
N10_neutral_neutral_022	15	0.663	0%	1.5, single single	49.74
N12_neutral_neutral_001	13	0.615	38%	delocalized	-147.92
N12_neutral_neutral_002	13	0.700	62%	1.5, single delocalized	-134.68
N12_neutral_neutral_003	13	0.801	69%	1.5, single delocalized	-112.01
N12_neutral_neutral_004	13	0.695	38%	1.5, double, single delocalized	-102.34
N12_neutral_neutral_005	14	0.760	29%	1.5, single double, single	-79.39
N12_neutral_neutral_006	14	0.722	29%	delocalized	-71.56
N12_neutral_neutral_007	18	0.645	0%	1.5, double, single single	-46.96
N12_neutral_neutral_008	15	0.787	20%	delocalized	-14.76
N12_neutral_neutral_009	18	0.605	0%	1.5, single single	94.51
N14_neutral_neutral_001	16	0.599	50%	delocalized	-140.21
N14_neutral_neutral_003	17	0.702	24%	1.5, single delocalized	-62.32
N14_neutral_neutral_004	10	1.480	80%	1.5, double, single delocalized	-38.65
N14_neutral_neutral_005	21	0.681	0%	1.5, double, long/weak, single, triple long/weak, single	-6.87
N4_neutral_neutral_001	6	0.796	0%	single	27.32
N4_neutral_neutral_002	4	1.118	50%	delocalized	53.79
N4_neutral_neutral_003	5	0.841	0%	1.5, double, single long/weak, single	58.33
N4_neutral_neutral_004	4	1.056	50%	delocalized	68.96
N6_neutral_neutral_001	5	1.248	80%	1.5, single delocalized	-63.71
N6_neutral_neutral_002	6	0.835	100%	1.5, single, triple delocalized	-39.62
				1.5	

(suite)

Nom	liaisons	BO Mayer moy.	% courtes	classes présentes	ΔH_f^{xtb}
N6_neutral_neutral_003	6	1.112	50%	delocalized 1.5, double, single, triple	-8.71
N6_neutral_neutral_004	3	2.336	100%	triple	-3.12
N6_neutral_neutral_005	6	1.071	67%	delocalized 1.5, long/weak, triple	0.91
N6_neutral_neutral_006	6	1.133	67%	delocalized 1.5, long/weak, single, triple	15.09
N6_neutral_neutral_007	7	0.867	29%	delocalized 1.5, single	19.92
N6_neutral_neutral_008	8	0.790	12%	double, single	20.81
N6_neutral_neutral_009	7	0.979	29%	double, single	47.09
N6_neutral_neutral_010	6	0.935	33%	single, triple	50.62
N6_neutral_neutral_012	9	0.729	0%	single	72.95
N8_neutral_neutral_001	9	0.698	56%	delocalized 1.5, single	-113.38
N8_neutral_neutral_002	8	0.891	75%	delocalized 1.5, single, triple	-103.47
N8_neutral_neutral_003	7	1.024	71%	delocalized 1.5, single, triple	-93.12
N8_neutral_neutral_005	7	1.158	71%	delocalized 1.5, single, triple	-83.97
N8_neutral_neutral_009	9	0.806	33%	double, single	-39.68
N8_neutral_neutral_010	8	0.910	50%	delocalized 1.5, single, triple	-29.32
N8_neutral_neutral_016	11	0.743	9%	delocalized 1.5, single	19.4
N8_neutral_neutral_019	10	0.828	20%	double, single	36.92
N8_neutral_neutral_024	11	0.618	18%	delocalized 1.5, long/weak, single	69.67

8. Confrontation à la littérature et aux bases externes

Le fichier `biblio_polyN/comparison_table_3sources.csv` de polyN-pipeline catalogue 337 structures connues (articles publiés + bases externes `N_csp` et `polyN_study`, relaxées au même

¹Triple, double ou délocalisée (ordre $\sim 1,5$), à l'exclusion des liaisons simples et des contacts non-liants.

niveau GFN2-xTB). Le croisement par (N, charge) avec nos 193 candidats donne trois résultats complémentaires :

- **Couverture** : 12 des 19 groupes (89 des 193 structures, 46 %) n'ont **aucune** référence cataloguée à cette formule/charge.
- **Topologie** : pour les 104 candidats des 7 groupes comparables, une empreinte de graphe de liaisons (séquence de degrés + tailles de cycles + nombre de liaisons, seuil 1,90 Å, NetworkX) montre que 76 (73 %) ne correspondent à aucune connectivité cataloguée. Au total, **165 des 193 candidats (86 %)** sont donc distincts de tout ce qui existe dans le tableau à 3 sources.
- **Stabilité** : 3 des 7 groupes comparables voient leur meilleur candidat local dépasser en stabilité le meilleur catalogué – N_9^- (−0,166 eV/atome), N_{10} (−0,066, malgré 60 références disponibles), N_{12} (−0,246). N_6 et N_8 retombent exactement sur le minimum déjà connu. N_4 est la seule perte nette (+0,302 eV/atome). **Correction post-publication** : une première version de ce rapport créditait aussi N_9^+ (−0,213 eV/atome) – ce candidat (N9_cation_cation_001) s'est avéré, après vérification de la connectivité (§2), être **fragmenté en trois espèces séparées (2+2+5 atomes)**, pas un cluster N_9^+ lié – retiré du classement. C'est actuellement le seul candidat N_9^+ terminé ; aucune comparaison de stabilité n'est donc possible pour ce groupe tant qu'un candidat non-fragmenté n'a pas fini.

Détail complet et graphique interactif : voir l'artefact [Uncharted Compositions](#).

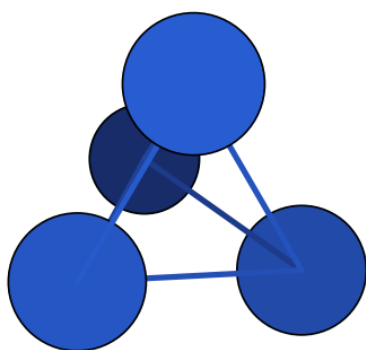
9. Limites et prochaines étapes

- La comparaison de stabilité de la section précédente reste au niveau **GFN2-xTB** – la confirmation définitive viendra des énergies WB97X-D4 elles-mêmes, une fois la campagne complète et les 19 candidats flaggés re-optimisés en vrais minima.
- Charges de Bader (QTAIM) et de Manz (DDEC6) reportées à une étape ultérieure – nécessitent respectivement le code de Henkelman et Chargemol sur un cube de densité via `orca_plot`, non installés sur le cluster à ce jour.
- Tous les fichiers ORCA bruts (`.gbw`, `.densities`, `.hess`) sont conservés intacts pour une seconde campagne CCSD(T) envisagée sur les petites structures – le module `refinement/` du dépôt compagnon `polyN` implémente déjà un enchaînement de méthodes config-driven avec seuil de taille (`max_n_atoms`) adapté à cet usage.
- Seuils de classification des liaisons N–N (§6) : triple < 1,15 Å, double < 1,22, délocalisée < 1,32, simple < 1,55, au-delà non-liant – purement géométriques, à recouper avec l'ordre de Mayer (fourni en regard dans `results_bonds.csv`) plutôt qu'à prendre isolément.

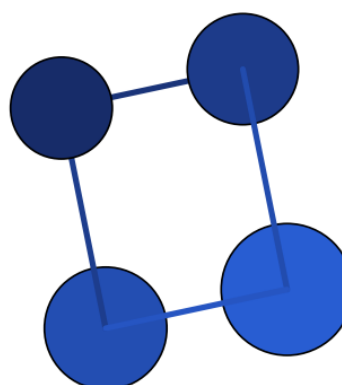
A. Structures optimisées (DFT, minima confirmés)

Les 90 candidats confirmés comme vrais minima au moment de la rédaction (aucune fréquence imaginaire – y compris, le cas échéant, des candidats fragmentés en plusieurs espèces, chaque fragment étant lui-même un minimum), deux par ligne. Trois figures, classées par nombre d'atomes croissant puis par stabilité au sein de chaque formule : Figure S1 (neutres), Figure S2 (cations), Figure S3 (anions). Chaque structure porte son nom, son groupe ponctuel, son ΔH (kcal/mol, isomère le plus stable *non-fragmenté* de sa formule = 0), le niveau de théorie de ce ΔH (DFT ou xTB si le DFT n'est pas encore terminé), sa référence (numéro(s) d'article ou *our work*), et la mention *fragmentée* avec la taille des morceaux si la structure n'est pas un cluster lié unique.

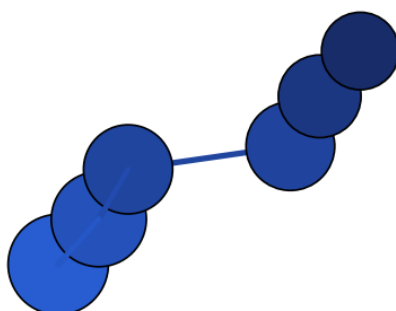
Figure S1 – composés neutres



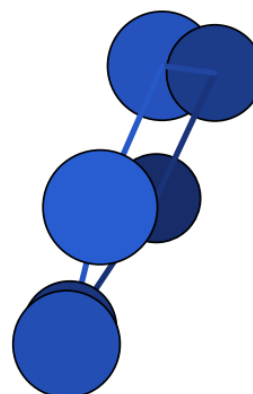
N4_neutral_neutral_001 – Td – $\Delta H = 0.00$
(DFT) – our work



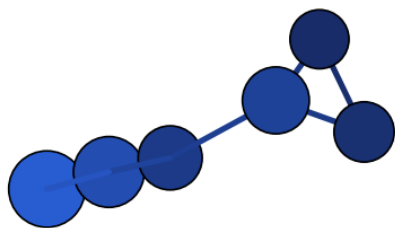
N4_neutral_neutral_004 – D2h – $\Delta H = 3.38$
(DFT) – our work



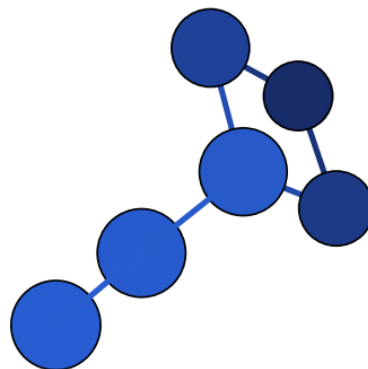
N6_neutral_neutral_001 – C2 – $\Delta H = 0.00$
(DFT) – our work



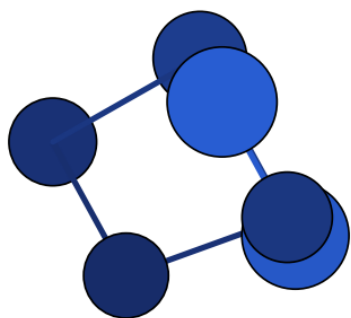
N6_neutral_neutral_002 – D2 – $\Delta H = 23.07$
(DFT) – [1], our work



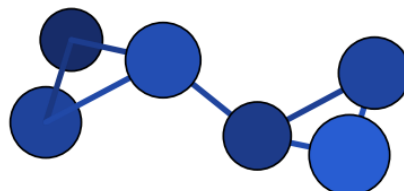
N6_neutral_neutral_003 – C1 – $\Delta H = 28.07$
(DFT) – our work



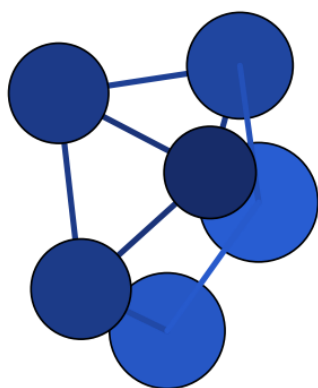
N6_neutral_neutral_005 – C1 – $\Delta H = 52.69$
(DFT) – our work



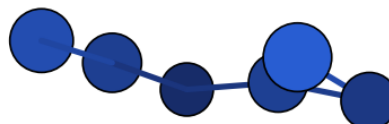
N6_neutral_neutral_007 – C2 – $\Delta H = 53.36$
(DFT) – our work



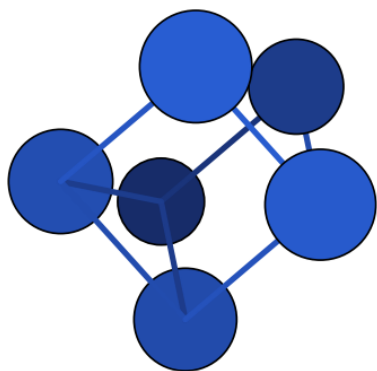
N6_neutral_neutral_009 – C2 – $\Delta H = 58.25$
(DFT) – our work



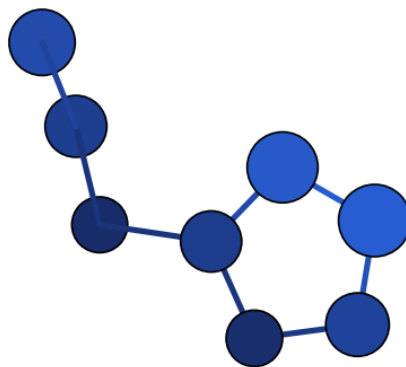
N6_neutral_neutral_008 – C2v – $\Delta H = 63.90$
(DFT) – our work



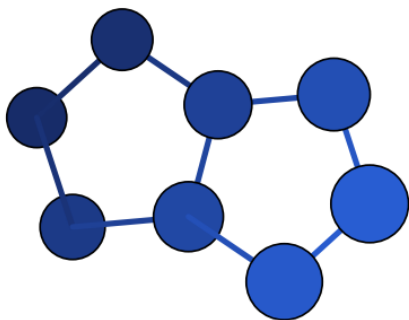
N6_neutral_neutral_006 – C1 – $\Delta H = 85.49$
(DFT) – our work



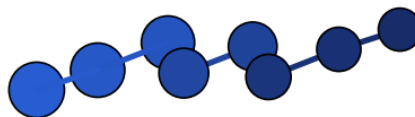
N6_neutral_neutral_012 – D3h –
 $\Delta H = 132.46$ (DFT) – our work



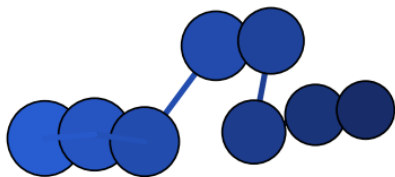
N8_neutral_neutral_002 – C1 – $\Delta H = 0.00$
(DFT) – [1],[2], our work



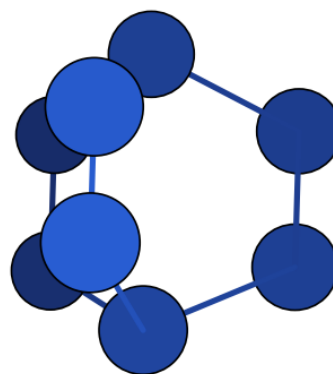
N8_neutral_neutral_001 – D2h – $\Delta H = 11.23$
(DFT) – [1],[2], our work



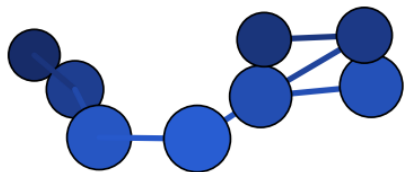
N8_neutral_neutral_005 – C2 – $\Delta H = 18.41$
(DFT) – [1], our work



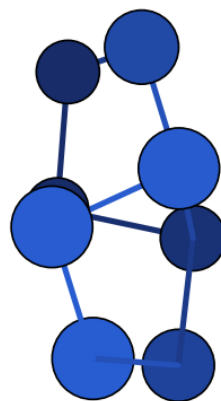
N8_neutral_neutral_003 – C1 – $\Delta H = 18.53$
(DFT) – [1], our work



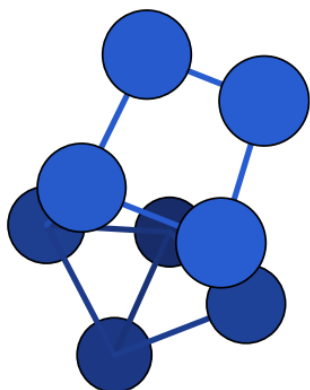
N8_neutral_neutral_009 – D3h – $\Delta H = 68.56$
(DFT) – our work



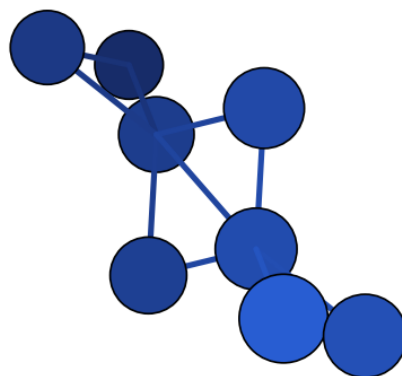
N8_neutral_neutral_010 – Cs – $\Delta H = 76.55$
(DFT) – our work



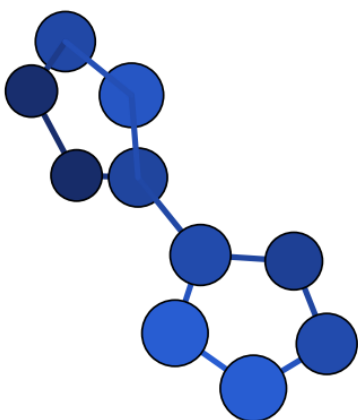
N8_neutral_neutral_019 – D2d –
 $\Delta H = 114.22$ (DFT) – our work



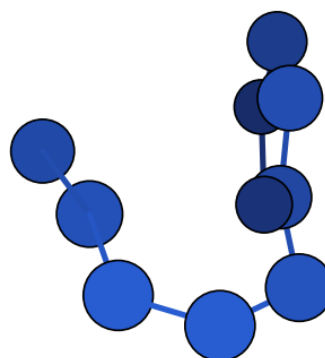
N8_neutral_neutral_016 – Cs – $\Delta H = 122.30$
(DFT) – our work



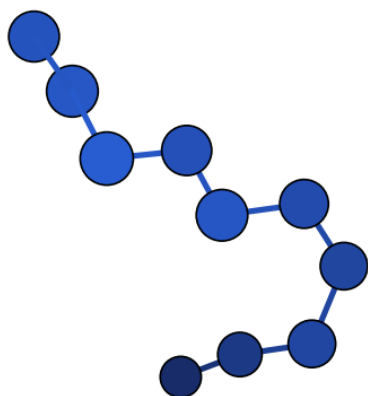
N8_neutral_neutral_024 – D2h –
 $\Delta H = 178.42$ (DFT) – our work



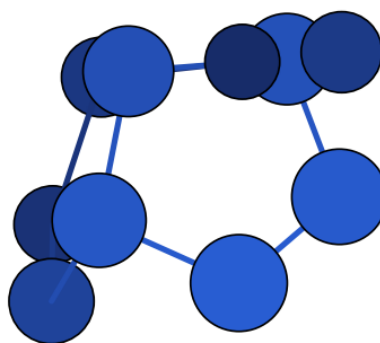
N10_neutral_neutral_001 – C2 – $\Delta H = 0.00$
(DFT) – our work



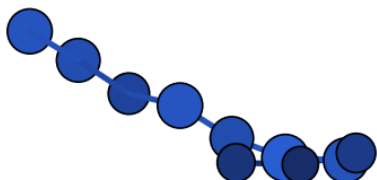
N10_neutral_neutral_002 – C1 – $\Delta H = 30.61$
(DFT) – our work



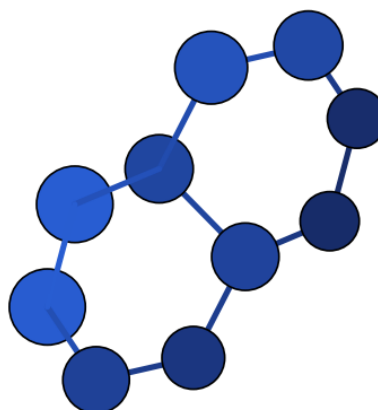
N10_neutral_neutral_003 – C1 – $\Delta H = 50.60$
(DFT) – [1], our work



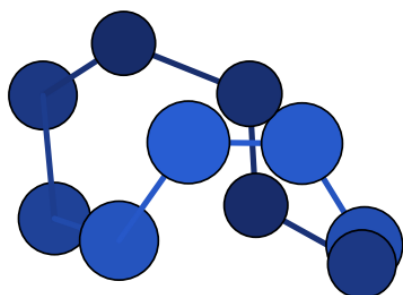
N10_neutral_neutral_006 – Cs – $\Delta H = 69.56$
(DFT) – our work



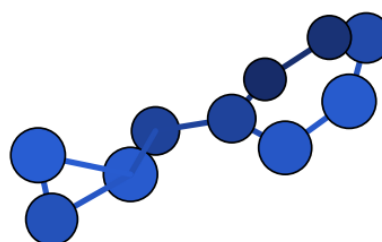
N10_neutral_neutral_004 – C1 – $\Delta H = 71.13$
(DFT) – [1], our work



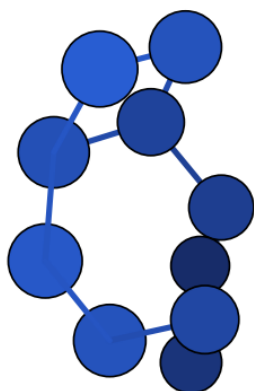
N10_neutral_neutral_007 – Cs – $\Delta H = 87.34$
(DFT) – our work



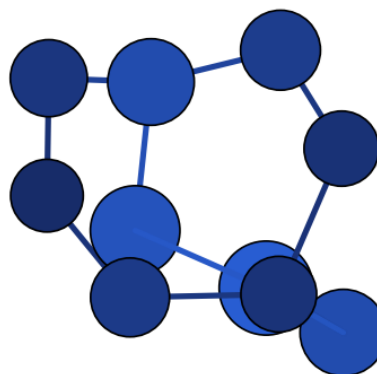
N10_neutral_neutral_005 – C1 –
 $\Delta H = 102.77$ (DFT) – our work



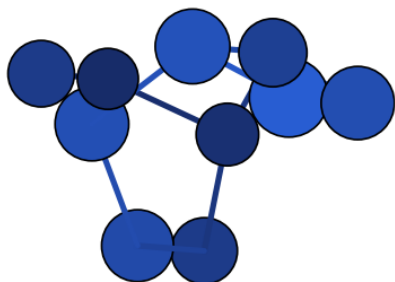
N10_neutral_neutral_011 – C1 –
 $\Delta H = 107.23$ (DFT) – our work



N10_neutral_neutral_010 – C1 –
 $\Delta H = 118.99$ (DFT) – our work



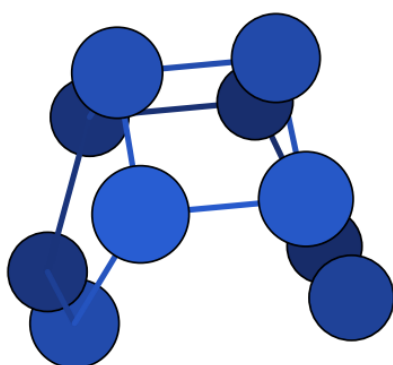
N10_neutral_neutral_015 – C2 –
 $\Delta H = 123.40$ (DFT) – our work



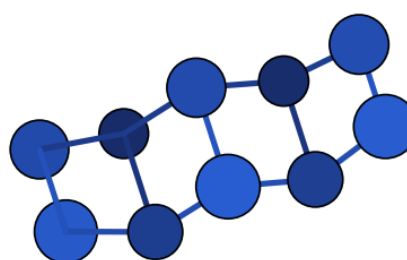
N10_neutral_neutral_016 – C1 –
 $\Delta H = 126.63$ (DFT) – our work



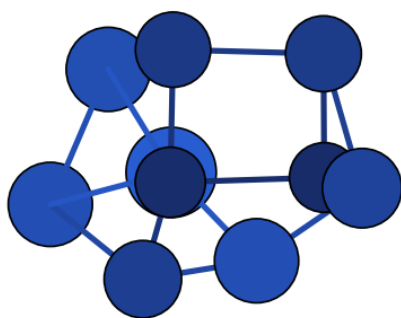
N10_neutral_neutral_014 – C1 –
 $\Delta H = 139.29$ (DFT) – our work



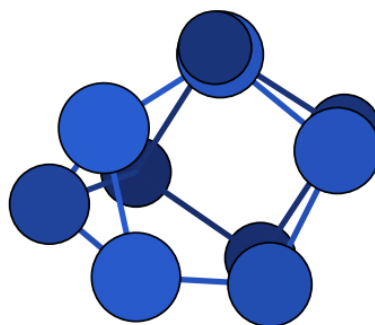
N10_neutral_neutral_017 – C2v –
 $\Delta H = 151.14$ (DFT) – our work



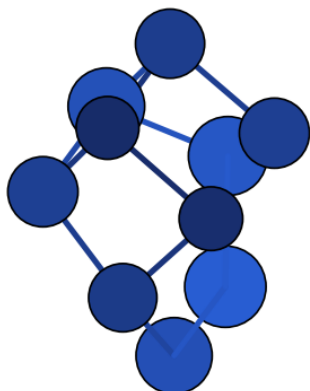
N10_neutral_neutral_021 – C2v –
 $\Delta H = 173.84$ (DFT) – our work



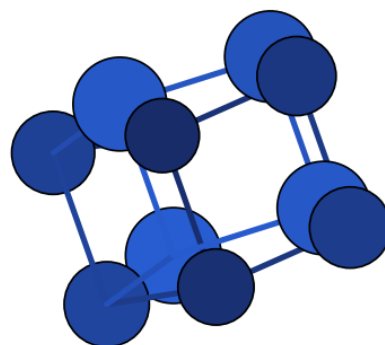
N10_neutral_neutral_018 – C2 –
 $\Delta H = 199.07$ (DFT) – our work



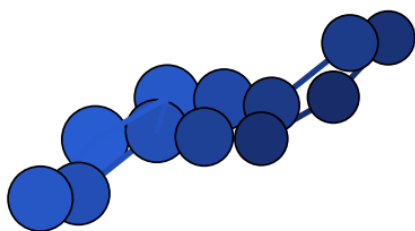
N10_neutral_neutral_019 – C3v –
 $\Delta H = 203.80$ (DFT) – our work



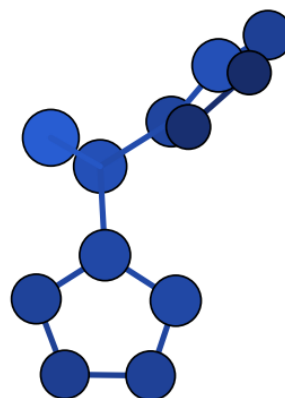
N10_neutral_neutral_020 – C2 –
 $\Delta H = 214.21$ (DFT) – our work



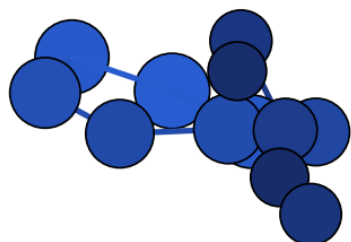
N10_neutral_neutral_022 – D5h –
 $\Delta H = 241.14$ (DFT) – our work



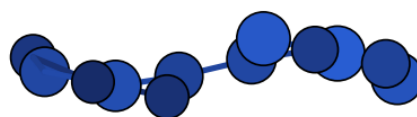
N12_neutral_neutral_001 – C1 – $\Delta H = 0.00$
(DFT) – our work



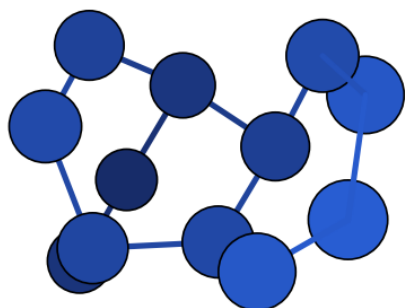
N12_neutral_neutral_003 – C1 – $\Delta H = 36.41$
(DFT) – our work



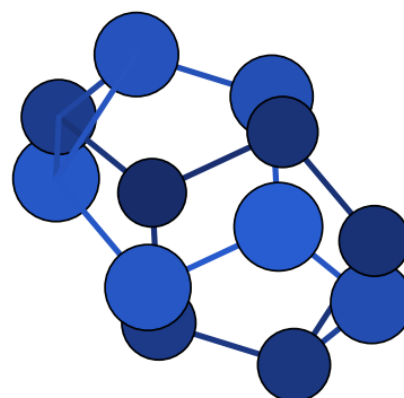
N12_neutral_neutral_002 – C1 – $\Delta H = 56.99$
(DFT) – our work



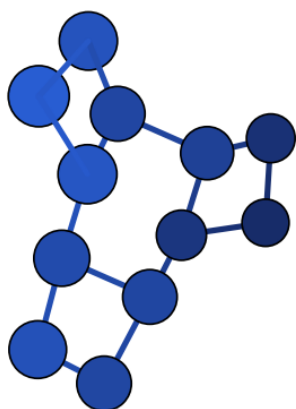
N12_neutral_neutral_004 – Cs – $\Delta H = 116.12$ (DFT) – our work



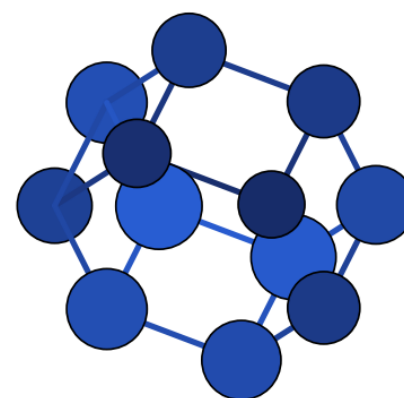
N12_neutral_neutral_006 – C1 – $\Delta H = 127.50$ (DFT) – our work



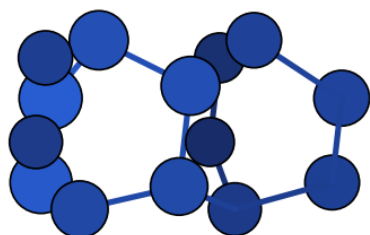
N12_neutral_neutral_007 – D3d – $\Delta H = 188.33$ (DFT) – our work



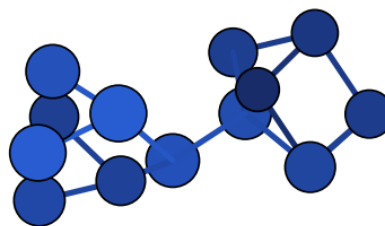
N12_neutral_neutral_008 – C1 – $\Delta H = 219.30$ (DFT) – our work



N12_neutral_neutral_009 – D6h – $\Delta H = 334.92$ (DFT) – our work

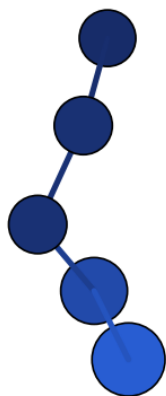


N14_neutral_neutral_003 – C2v –
 $\Delta H = 17.64$ (DFT) – our work

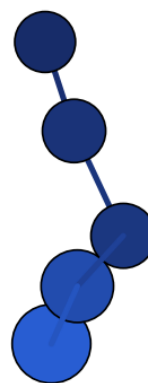


N14_neutral_neutral_005 – C2 – $\Delta H = 90.77$
(DFT) – our work

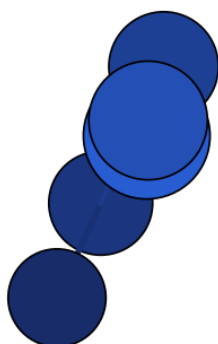
Figure S2 – composés cationiques



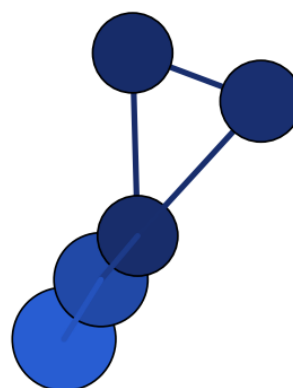
N5_cation_cation_003 – C1 – $\Delta H = 0.00$
(DFT) – our work



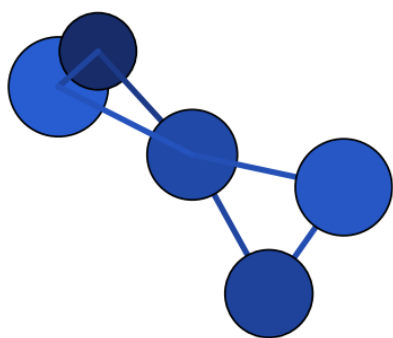
N5_cation_cation_001 – C2v – $\Delta H = 0.00$
(DFT) – our work



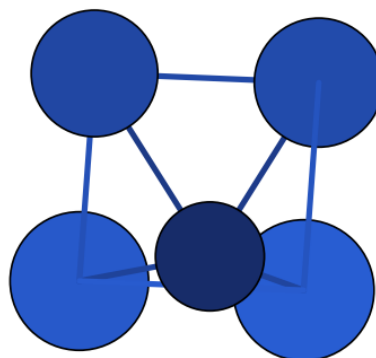
N5_cation_cation_004 – C1 – $\Delta H = 0.01$
(DFT) – our work



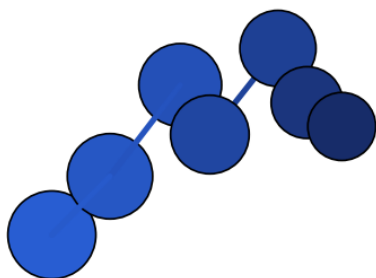
N5_cation_cation_002 – Cs – $\Delta H = 41.63$
(DFT) – our work



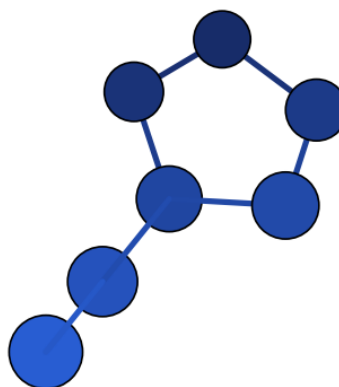
N5_cation_cation_005 – D2d – $\Delta H = 109.53$
(DFT) – our work



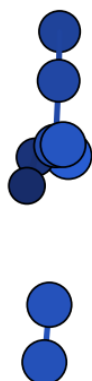
N5_cation_cation_006 – C4v – $\Delta H = 155.74$
(DFT) – our work



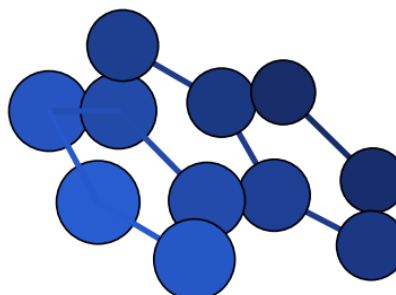
N7_cation_cation_001 – Cs – $\Delta H = 0.00$
(DFT) – our work



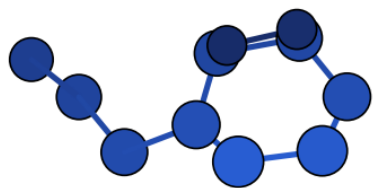
N7_cation_cation_003 – Cs – $\Delta H = 25.42$
(DFT) – our work



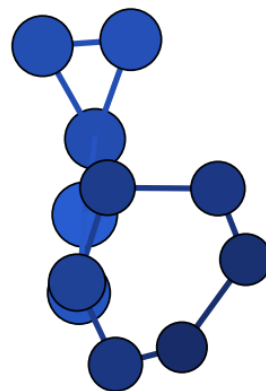
N9_cation_cation_001 – C1 – $\Delta H = -29.44$
(xtb) – our work – **fragmentée (2+2+5)**



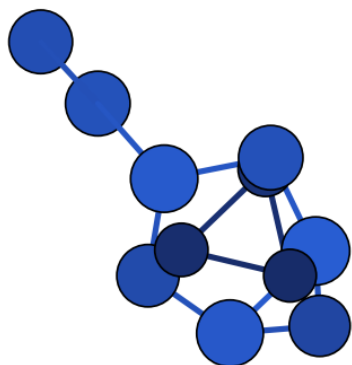
N11_cation_cation_001 – C2v – $\Delta H = 23.30$
(DFT) – our work



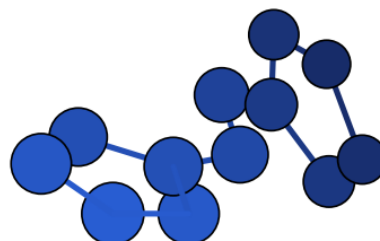
N11_cation_cation_004 – C1 – $\Delta H = 56.05$
(DFT) – our work



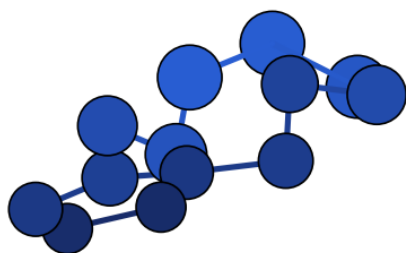
N11_cation_cation_006 – C1 – $\Delta H = 67.68$
(DFT) – our work



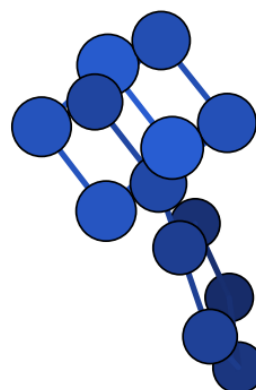
N11_cation_cation_007 – C1 – $\Delta H = 118.69$
(DFT) – our work



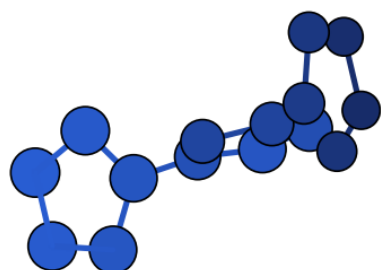
N12_cation_cation_001 – C1 – $\Delta H = 0.00$
(DFT) – our work



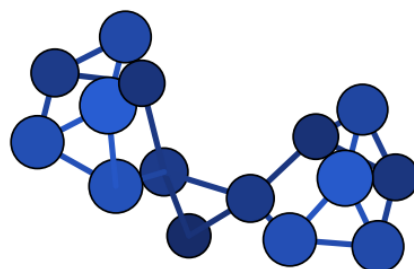
N13_cation_cation_001 – C1 – $\Delta H = 0.00$
(DFT) – our work



N13_cation_cation_004 – C1 – $\Delta H = 151.22$
(DFT) – our work

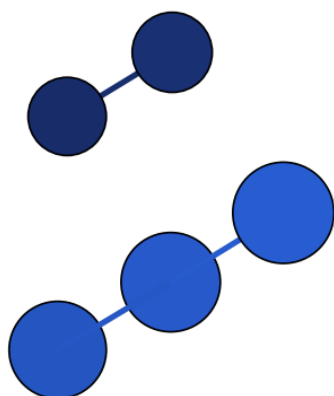


N15_cation_cation_001 – C1 – $\Delta H = 0.00$
(DFT) – our work

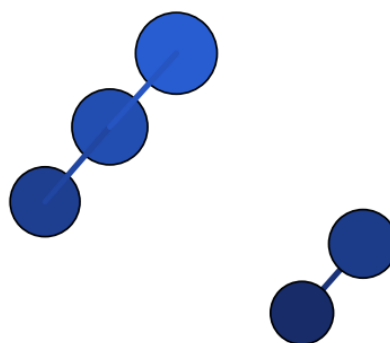


N15_cation_cation_002 – C1 – $\Delta H = 327.87$
(DFT) – our work

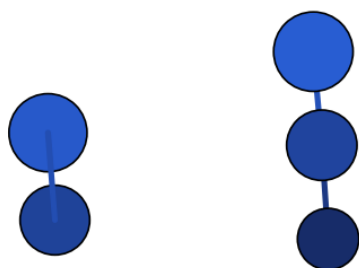
Figure S3 – composés anioniques



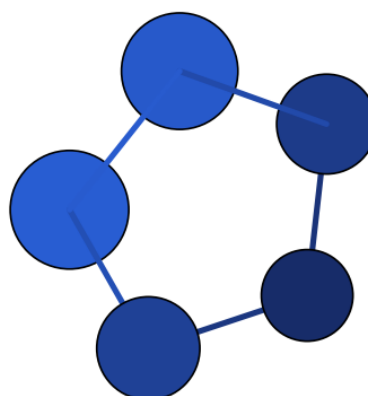
N5_anion_anion_002 – C1 – $\Delta H = -6.58$
(DFT) – our work – **fragmentée (2+3)**



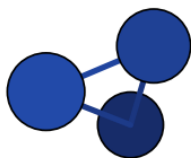
N5_anion_anion_003 – C1 – $\Delta H = -6.58$
(DFT) – our work – **fragmentée (2+3)**



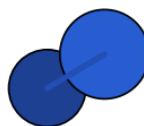
N5_anion_anion_004 – C1 – $\Delta H = -6.58$
(DFT) – our work – **fragmentée (2+3)**



N5_anion_anion_001 – D5h – $\Delta H = 0.00$
(DFT) – our work



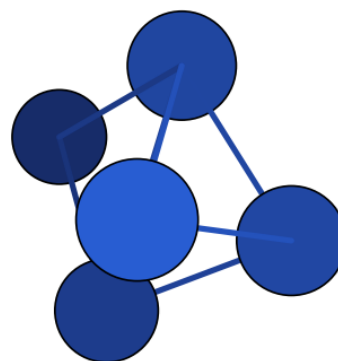
N5_anion_anion_007 – C1 – $\Delta H = 63.83$
(DFT) – our work – **fragmentée (2+3)**



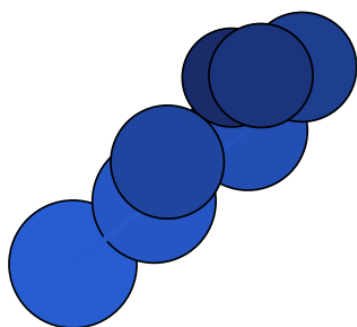
N5_anion_anion_005 – C1 – $\Delta H = 63.83$
(DFT) – our work – **fragmentée (2+3)**



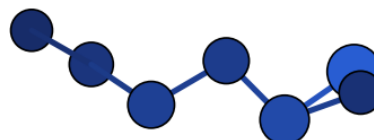
N5_anion_anion_008 – C1 – $\Delta H = 65.57$
(DFT) – our work – **fragmentée (2+3)**



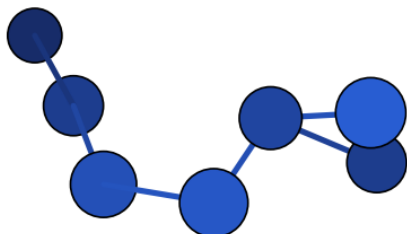
N5_anion_anion_010 – C2v – $\Delta H = 170.76$
(DFT) – our work



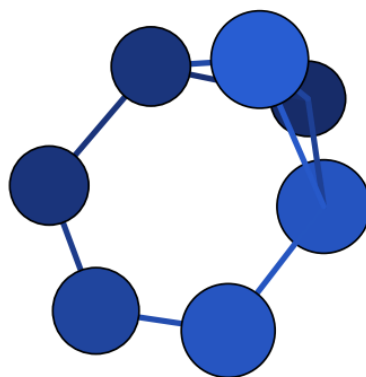
N7_anion_anion_001 – C1 – $\Delta H = 0.00$ (DFT)
– our work



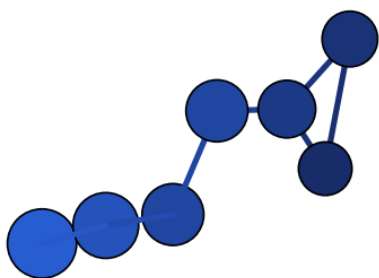
N7_anion_anion_011 – C1 – $\Delta H = 26.27$
(DFT) – our work



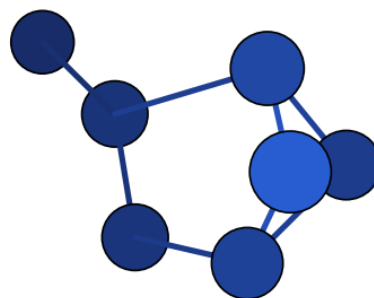
N7_anion_anion_012 – C1 – $\Delta H = 29.35$
(DFT) – our work



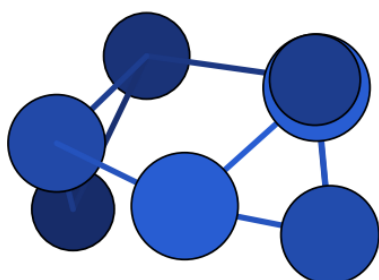
N7_anion_anion_010 – C2v – $\Delta H = 46.25$
(DFT) – our work



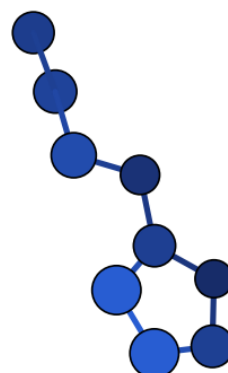
N7_anion_anion_008 – C1 – $\Delta H = 49.34$
(DFT) – our work



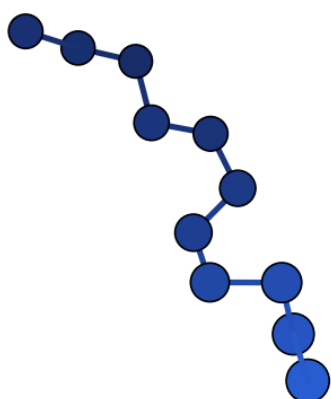
N7_anion_anion_014 – Cs – $\Delta H = 55.87$
(DFT) – our work



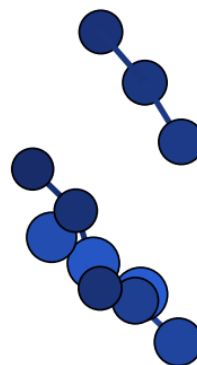
N7_anion_anion_019 – C1 – $\Delta H = 89.71$
(DFT) – our work



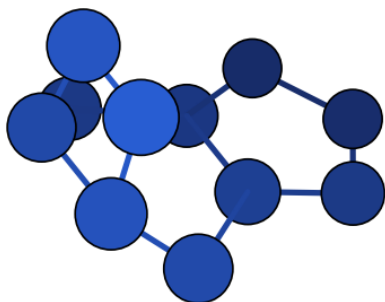
N9_anion_anion_001 – C1 – $\Delta H = 0.00$ (DFT)
– our work



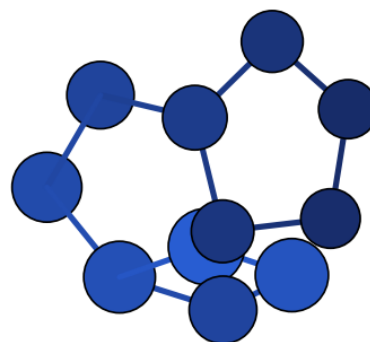
N11_anion_anion_006 – C1 – $\Delta H = 60.32$
(DFT) – our work



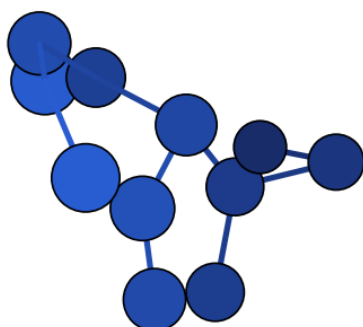
N11_anion_anion_011 – C1 – $\Delta H = 74.02$
(DFT) – our work – **fragmentée (3+3+5)**



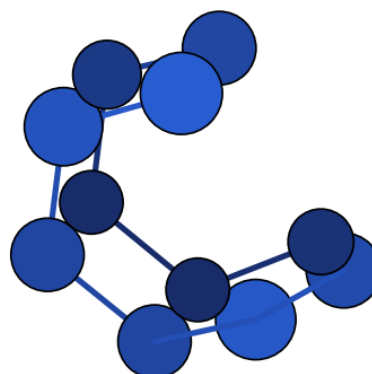
N11_anion_anion_007 – Cs – $\Delta H = 101.73$
(DFT) – our work



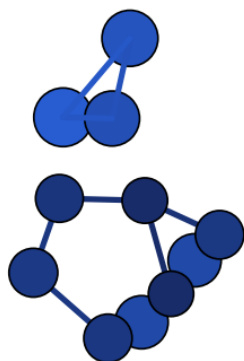
N11_anion_anion_008 – C1 – $\Delta H = 108.61$
(DFT) – our work



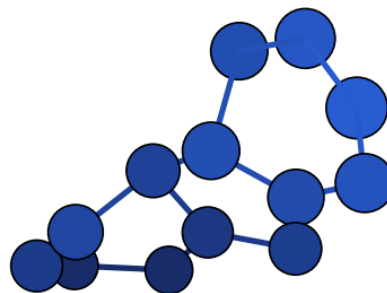
N11_anion_anion_013 – C1 – $\Delta H = 177.01$
(DFT) – our work



N11_anion_anion_017 – C1 – $\Delta H = 185.47$
(DFT) – our work



N11_anion_anion_018 – C1 – $\Delta H = 199.63$
(DFT) – our work – **fragmentée (3+8)**



N13_anion_anion_001 – C1 – $\Delta H = 0.00$
(DFT) – our work

B. Références

Articles cités dans les tableaux de la section 2 (numérotation propre à ce rapport ; correspondance avec les codes de `biblio_polyN/table_references.tex` de `polyN-pipeline` en annotation).

- [1]] Fau, Mobita, Wilson, Perera, Bartlett, *Quantum Theory Project report*, University of Florida.
- [2]] Glukhovtsev, Jiao, von Ragé Schleyer, *Inorg. Chem.* **1996**, 35, 7124–7133.